Operations Research (OR) TD 4

Simplex + Graphical sensitivity — 3RD YEAR - ENSIA

Exercice C1

Show using simplex that the following LP's are unbounded:

Max $z = x_1 + 2x_2$

S.t. $-x_1 + x_2 \leq 2$

$-2x_1 + x_2 \leq 1$

$x_1, x_2 \geq 0$

Min $z = -x_1 - 3x_2$

S.t. $x_1 - 2x_2 \leq 4$

$-x_1 + x_2 \leq 3$

$x_1, x_2 \geq 0$

Exercice C2

Max $z = 5x_1 + 3x_2$

S.t. $4x_1 + 2x_2 \leq 12$

$4x_1 + x_2 \leq 10$

$x_1 + x_2 \leq 4$

$x_1, x_2 \geq 0$

(a) Show that this LP exhibit degeneracy

(b) Graph the feasible region and show which corner correspond to this degeneracy

Exercice C3

Consider the following two-variable linear programming problem:

Max $z = C_1 x_1 + C_2 x_2$

S.t. $a_{11} x_1 + a_{12} x_2 \leq b_1$

$a_{21} x_1 + a_{22} x_2 \leq b_2$

$x_1, x_2 \geq 0$

(a) Prove that, in the simplex method, the minimum ratio test must be satisfied when selecting the leaving variable to avoid infeasible solutions.

Exercice C4

Radioco manufactures two types of radios. The only scarce resource that is needed to produce radios is labor. At present, the company has two laborers. Laborer 1 is willing to work up to 40 hours per week and is paid $5 per hour. Laborer 2 will work up to 50 hours per week for $6 per hour. The price as well as the resources required to build each type of radio are given in Table below. Letting x_i be the number of Type i radios produced each week, Radioco should solve the following LP:

Max $z = 3x_1 + 2x_2$

S.t. $x_1 + 2x_2 \leq 40$

$2x_1 + x_2 \leq 50$

$x_1, x_2 \geq 0$

(a) For what values of the price of a Type 1 radio would the current BFS remain optimal?

(b) For what values of the price of a Type 2 radio would the current BFS remain optimal?

(c) If laborer 1 were willing to work only 30 hours per week, then would the current BFS remain optimal? Find the new optimal solution to the LP.

(d) If laborer 2 were willing to work up to 60 hours per week, then would the current BFS remain optimal? Find the new optimal solution to the LP.

(e) Find the shadow price of each constraint.

Radio 1		**Radio 2**	
Price ($)	**Resource required**	**Price ($)**	**Resource required**
25	Laborer 1: 1 hour	22	Laborer 1: 2 hours
	Laborer 2: 2 hours		Laborer 2: 2 hours
	Raw material cost: $5		Raw material cost: $4

Exercice P1 (Optional)

Find all optimal solutions to the following LP:

Max $z = 3x_1 + 3x_2$

S.t. $x_1 + x_2 \leq 1$

$x_1, x_2 \geq 0$

Exercice P2 (Optional)

Find alternative optimal solutions to the following LP:

Max $z = x_1 + x_2$

S.t. $x_1 + x_2 + x_3 \leq 1$

$x_1 + \quad + 2x_3 \leq 1$

$x_1, x_2, x_3 \geq 0$

Exercice P3 (Optional)

Explain why the Phase I of Two-Phase method LP will usually have alternative optimal solutions.

Exercice P4 (Optional)

Wild West produces two types of cowboy hats. A Type 1 hat requires twice as much labor time as a Type 2. If all the available labor time is dedicated to Type 2 alone, the company can produce a total of 400 Type 2 hats a day. The respective market limits for the two types are 150 and 200 hats per day. The revenue is $8 per Type 1 hat and $5 per Type 2 hat.

(a) Use the graphical solution to determine the number of hats of each type that maximizes revenue.

(b) Determine the dual price of the production capacity (in terms of the Type 2 hat) and the range for which it is applicable.

(c) If the daily demand limit on the Type 1 hat is decreased to 120, use the dual price to determine the corresponding effect on the optimal revenue.

(d) What is the dual price of the market share of the Type 2 hat? By how much can the market share be increased while yielding the computed worth per unit?

Exercice P5 (Optional)

Show that the following LP is infeasible.

$$
\begin{aligned}
\textbf{Max } z = {} & x_1 + x_2 \\
\textbf{S.t. } & x_1 + 2x_2 \leq 3 \\
& 2x_1 + x_2 \geq 8 \\
& x_1 , x_2 \geq 0
\end{aligned}
$$